



METALS, MINERALS & ORES — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

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1. Metallurgy, Metallic Character & Reactivity

- Smelting mein ore ko high temperature par reduce karke metal molten/fused state mein obtain kiya jata hai. Roasting, calcination aur froth-floatation alag metallurgical operations hain. [Q1]
- Electronic configuration 2,8,8,2 Calcium ($Z=20$) ki hai; outer shell ke 2 electrons lose karke Ca^{2+} banana metallic/electropositive behaviour dikhata hai. [Q2]
- Electrophile electron-pair acceptor hota hai. BF_3 aur H^+ strongly electron-deficient electrophiles hain; source Na^+ ko stable-octet ke karan “true electrophile” nahi maanta. [Q3]
- Common reactivity series: $\text{K} > \text{Na} > \text{Ca} > \text{Mg} > \text{Al} > \text{Zn} > \text{Fe} > \text{Pb} > \text{H} > \text{Cu} > \text{Hg} > \text{Ag} > \text{Au}$. Given options mein Potassium sabse reactive metal hai. [Q4]
- Food cans par Tin coating Zinc se safer hoti hai, kyunki Zn zyada reactive hai aur food acids ke saath react kar sakta hai; Sn comparatively less reactive hai. [Q5]

2. Magnetism, Hardness, Ductility & Extreme Properties

- Platinum paramagnetic hai; Iron/Nickel ferromagnetic aur Mercury diamagnetic behaviour dikhata hai. Cu^{2+} paramagnetic aur Gd ferromagnetic pair hai. [Q6, Q7]
- Lithium sabse light metal (very low density) aur Osmium densest naturally occurring metal maana jata hai. [Q8]
- Given metals mein Tungsten hardest hai; natural substances mein Diamond conventionally hardest maana jata hai. [Q9, Q10, Q11]
- Nichrome (mainly Ni + Cr, often Fe) hard, ductile, high-resistivity aur high-temperature resistant alloy hai; heating elements mein useful hota hai. [Q12]

3. Heavy / Costly Metals — Meaning Matters

- Uranium ($Z=92$) heaviest naturally occurring element ke standard exam fact ke roop mein poochha jata hai; Uranium-238 ka nucleus 92 protons + 146 neutrons rakhta hai. [Q13, Q14]
- Given options Gold, Silver, Mercury aur Platinum mein density ke basis par Platinum sabse heavy hai ($\sim 21.1 \text{ g/cm}^3$). [Q15]



EXAM TRAP — “Heaviest” ka matlab clear karo

- Densest metal = Osmium; “heaviest natural element” = Uranium (atomic-number/mass context); kisi option-set mein “heaviest metal” density ke hisab se Platinum ho sakta hai. Wording dekhe bina ek hi fact apply mat karo. [Q8, Q13–Q15]

- Q16 “recently discovered world’s costliest metal” ambiguous hai: Endohedral fullerene metal nahi aur Californium-252 1950 mein discover hua tha. Source ne answer disputed (*) rakha hai. [Q16]

⚠ EXAM TRAP — Q16 — Costliest Metal Question Disputed

► Price-per-gram lists time/source ke saath badalti hain, aur “recently discovered” phrase bhi vague hai. Is question ko fixed one-line fact ke roop mein memorize mat karo. [Q16]

4. Pearl, Ruby, Sapphire & Corundum

- Pearl ka major mineral component Calcium Carbonate hota hai, mostly Aragonite form mein (~82–86%); Conchiolin protein binding matrix ka kaam karta hai. [Q17, Q18, Q19]
- Ruby aur Sapphire dono gem-quality Corundum hain; chemically Corundum = Aluminium Oxide, Al_2O_3 . Trace impurities different colours produce karti hain. [Q20, Q21]

5. Sodium, Noble Metals, Gold & White Metals

- Sodium air/moisture ke saath vigorously react karta hai, isliye kerosene ke andar store kiya jata hai. Air mein “sif melt” hona reason nahi; major issue rapid/exothermic reaction hai. [Q22, Q23]
- Sodium-vapour street lamps characteristic yellow/orange light deti hain; low-pressure sodium light nearly monochromatic hoti hai, isliye fog/rain context mein visibility-oriented exam questions mein poochhi jati hai. [Q24, Q25]
- Gold very unreactive/noble metal hai aur native/free state mein mil sakta hai. Noble metals oxidation/corrosion resist karte hain; Uranium aur Lead noble metals nahi hain. [Q26, Q27]
- Pure Gold = 24 carat. Jewellery ke liye pure gold bahut soft hota hai, isliye Cu/Ag jaise metals ke saath alloy karke 22K/18K banaya jata hai. [Q29]
- “White metal” term broad/ambiguous hai; Platinum, Palladium aur Rhodium naturally silvery-white platinum-group metals hain, isliye source ne more-than-one option key kiya. [Q30]

⚠ EXAM TRAP — Gold kis acid mein dissolve hota hai?

► Gold kisi single H_2SO_4 , HNO_3 ya HCl mein normally dissolve nahi hota. Aqua regia = concentrated $\text{HCl} : \text{HNO}_3 \approx 3:1$ gold dissolve karti hai. Q28 ke options mein correct mixture tha hi nahi, isliye source answer *. [Q28]

6. Minerals, Talc, Quicklime, Limestone & Cement

- Minerals geologically naturally occurring inorganic solids hote hain jinki characteristic composition aur physical properties hoti hain. [Q31]
- Ilmenite (FeTiO_3) Titanium ka important ore/mineral hai aur Indian coastal beach sands mein milta hai. [Q32]
- Talc/soapstone hydrated Magnesium Silicate hai aur Mohs scale par hardness 1 ke saath softest standard mineral maana jata hai. [Q33]
- Quicklime = Calcium Oxide (CaO); Limestone mainly Calcium Carbonate (CaCO_3) hota hai. [Q34, Q35]
- Cement mein lime/limestone-derived CaO major constituent hota hai, lekin silica, alumina/other clay components aur gypsum bhi formulation ka part hote hain; source question ne “more than one” key kiya. [Q36]

⚠ EXAM TRAP — Cement — “Main Ingredient” vs “Ingredients”

► Ordinary Portland cement ka largest oxide fraction CaO hota hai, lekin cement single-substance material nahi. Question agar ingredients/options multiple de to wording aur official key ko carefully read karo. [Q36]

7. Gypsum & Plaster of Paris

- Gypsum = Calcium Sulphate Dihydrate, $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$. Isko controlled heating/calcination se Plaster of Paris banaya jata hai. [Q37, Q38, Q42]
- Plaster of Paris ka exact formula $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$ hai, yani Calcium Sulphate Hemihydrate. Doctors casts, artists moulds aur sculptors ise use karte hain. [Q39, Q40, Q41]

⚠ EXAM TRAP — POP ko sirf “Calcium Sulphate” bolna incomplete hai

► Family name calcium sulphate sahi hai, lekin exact compound = calcium sulphate hemihydrate ($\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$); Gypsum = dihydrate ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$). [Q39, Q40, Q42]

8. Monazite, Mica & Thermal Conductivity

- Monazite Thorium ka important ore hai; ismein rare-earth elements jaise Lanthanum/Cerium bhi ho sakte hain, aur Indian coastal sands monazite-rich areas ke liye known hain. [Q43]
- Mica electricity ka poor conductor/achha insulator hai, lekin heat tolerate/conduct kar sakta hai; isliye electrical/electronics industry mein insulating raw material ke roop mein use hota hai. [Q44, Q45]
- Given liquids Water/Ether/Benzene/Mercury mein Mercury metallic nature ki wajah se best heat conductor hai; metals mein bhi Mercury comparatively very poor thermal conductor hai. [Q46, Q47]

9. Liquid Metals, Mercury & Reaction with Water

- Mercury ordinary room temperature par liquid metal hai (quicksilver). Thermometers mein historically uniform expansion, visible column aur glass ko wet na karne jaise properties useful rahi hain. [Q48, Q50]
- Lead water ke saath generally react nahi karta; Copper steam ke saath bhi react nahi karta. Mg hot water aur Al/Fe steam ke saath reaction show kar sakte hain. [Q51, Q52]

⚠ EXAM TRAP — Q49 — Francium “room-temperature liquid”

► Standard room-temperature examples: Mercury (metal) aur Bromine (non-metal) liquid hote hain. Francium ka melting point estimate room temperature ke bahut paas aur experimentally uncertain hai; source ne given options mein Francium key kiya, ise exam-specific fact samjho. [Q49]

10. Iron Ores, Rusting & Galvanization

- Iron ka major ore Haematite (Fe_2O_3) hai; Bailadila (Chhattisgarh) high-grade Haematite iron ore ke liye famous hai. [Q53, Q54]
- Rusting mein Iron oxygen + moisture se hydrated iron(III) oxide banata hai; oxygen/moisture add hone ke karan object ka measured weight initially increase karta hai. [Q55, Q56, Q57]
- Iron corrosion/rusting ke liye oxygen aur moisture dono essential conditions hain. [Q62]
- Galvanization mein iron/steel par Zinc coating hoti hai. Zinc barrier bhi deta hai aur preferential oxidation/sacrificial protection ke through iron ko rusting se bachata hai. [Q58, Q59, Q60, Q61]

⚠ EXAM TRAP — “Galvanized iron does not rust” absolute statement nahi

► Intact zinc coating strong protection deti hai, lekin coating damage/exhaust hone par underlying iron rust kar sakta hai. Exam assertion ko galvanization principle ke context mein samjho. [Q58–Q61]

11. Protective Coatings & Reducing Power

- Brass/Copper utensils par Tin coating ko kalai kehte hain; ye food ko copper surface se react karke contamination/patina products banane se reduce karti hai. [Q63]
- Aluminium anodizing mein surface par thick protective Aluminium Oxide (Al_2O_3) layer develop ki jati hai, jo corrosion resistance aur durability improve karti hai. [Q64]
- Metal ka standard electrode potential jitna more negative, uski reducing power generally utni strong. E° : $\text{B} = -3 \text{ V}$, $\text{C} = -1.2 \text{ V}$, $\text{A} = +0.5 \text{ V} \Rightarrow \text{B} > \text{C} > \text{A}$. [Q65]

12. Key Metal Properties & Electrical Conductivity

- Key matching: Sodium soft hai aur knife se cut ho sakta hai; Mercury room temperature par liquid; Silver best electrical conductor; Lead poor heat conductor. [Q66]
- Exam properties mapping: best heat/electric conductor = Silver; Earth's crust mein most abundant metal = Aluminium; most malleable/flexible metal = Gold; low heat conductivity among listed metals = Lead. [Q67]
- Given materials mein Silver ki electrical conductivity highest hai; Diamond electrical insulator hai, Graphite conductive hai lekin Silver se kam. [Q68]

13. Aluminium — Bauxite and Other Ores

- Bauxite Aluminium ka principal/most important ore aur aluminium industry ka main raw material hai. Aluminium metal industrially bauxite-derived alumina se extract ki jati hai. [Q69, Q70, Q71, Q72, Q73, Q74]

- Aluminium ke important minerals/ores mein Bauxite, Corundum (Al_2O_3), Cryolite (Na_3AlF_6) aur Diaspore include kiye jate hain. [Q75]

14. Industrial Processes & Material–Composition Matching

- Calcium Carbonate toothpaste mein mild abrasive/filler ingredient ho sakta hai. Bordeaux mixture = Copper Sulphate + Lime; dental amalgam traditionally Hg + Ag/Sn/Cu alloy system hota hai. [Q76]

EXAM TRAP — Liquid Sodium Coolant — context specific

- Molten sodium Fast Breeder Reactors/fast reactors mein coolant ke roop mein use ho sakta hai, lekin “all nuclear reactors” ke liye general statement sahi nahi. Q76 mein isi over-generalization ki wajah se CaCO_3 -toothpaste statement correct key hai. [Q76]

- Industrial-process matching: Cracking–Petroleum; Smelting–Copper/metallurgy; Hydrogenation–Edible oils/fats; Vulcanization–Rubber. [Q77]
- Composition matching: Diamond–Carbon; Marble–Calcium (mainly CaCO_3); Sand–Silicon (mainly SiO_2); Ruby–Aluminium (Al_2O_3). [Q78]
- Ore matching: Bauxite–Aluminium, Calamine–Zinc, Haematite–Iron sahi hain; Cinnabar Mercury ka ore hai, Copper ka nahi. [Q79]

15. Density & Melting-Point Comparisons

- Sodium ki density water se kam ($\sim 0.97 \text{ g/cm}^3$), isliye water se lighter hai — though water ke saath violently react karta hai, so “float” demonstration unsafe hai. [Q80]
- Density decreasing order: Gold (~ 19.3) > Mercury (~ 13.6) > Steel ($\sim 7.8 \text{ g/cm}^3$). [Q81]
- Given substances Boron, Iron, Silicon, Aluminium mein melting point Boron ka highest hai; approximate order $\text{B} > \text{Fe} > \text{Si} > \text{Al}$. [Q82]